Is the Safety and Hallucination Feature Geometry of Language Models Conditional on Dialogue Context? A Confound-Controlled Study

Anonymous authors
Paper under double-blind review

Abstract

A growing class of inference-time alignment methods, conditional activation steering and conditional subspace disentanglement of hallucination from safety, rests on an implicit premise: that the low-dimensional residual-stream directions encoding safety/refusal and epistemic uncertainty are *conditional* on dialogue context, so that a single static intervention learned on one context is mis-specified for another. We ask whether this premise is empirically true, and find that the obvious ways of measuring it are confounded. We introduce a confound-controlled protocol: subspaces are recovered by contrastive difference-of-means, drift is measured as a gauge-invariant chordal distance between subspaces, and "the subspace moved with context" is tested against (i) a turn-1 baseline that removes the generic no-context→context jump, (ii) a length- and topic-matched *scrambled-twin* null that holds token count and lexical content fixed while destroying coherent meaning, and (iii) a dialogue-level bootstrap confidence interval so that no single dialogue can carry the result. Applied to an instruction-tuned model, three intuitive positive measurements (cumulative displacement, between-dialogue spread versus a same-context floor, and a small-sample placebo comparison) are each overturned by the next control. Under the full protocol, the apparent context-conditionality of the hallucination subspace is statistically indistinguishable from a length/topic artifact: scrambled twins move the subspace as much as real dialogues, and the real-versus-placebo confidence intervals overlap at every layer. We further fit the static disentangling projector these methods would replace and show its across-turn re-entanglement is fully matched by the length/topic placebo, i.e. the static baseline does not genuinely fail. Beyond the specific null, the transferable contributions are a reusable protocol, a taxonomy of three confounds that each independently produce a false positive (a no-context onset, a same-context null that is rigged to fire, and small-sample single-dialogue effects), and a control checklist for anyone making a "context changes the representation" claim; we also derive an actionable requirement for conditional-alignment methods, namely that they report this controlled measurement before assuming the premise. We report the null as scoped to the models and conditions tested. **All numbers in this draft are from real runs on the released code. The null replicates across two model scales in raw activation space (0.5B, 1.5B) and, critically, in sparse-autoencoder feature space (the space disentanglement methods operate in), where the scrambled twin moves the subspace *more* than real dialogues at the late layers. The larger-instruction-model and mixture-of-experts cells are in progress and are marked as such.**

1 Introduction

Mechanistic interpretability has established that high-level behaviors of large language models (LLMs) are often approximately linearly encoded: refusal is mediated by a single residual-stream direction (Arditi et al., 2024), truth/falsehood is linearly represented (Marks & Tegmark, 2023), and sparse autoencoders (SAEs) recover interpretable feature directions at scale (Bricken et al., 2023; Templeton et al., 2024; Lieberum et al., 2024). These directions can be read and steered (Zou et al., 2023; Li et al., 2023; Panickssery et al., 2024) and erased in closed form (Ravfogel et al., 2020; Belrose et al., 2023).

A natural next step, pursued by recent work, is to *disentangle* behaviors that share latent components: for example, to mitigate hallucination without degrading safety by orthogonalizing the SAE subspaces that encode them (Mahmoud et al., 2025). Once one moves from a single static edit to a *conditional* one, an implicit empirical premise enters: that the relevant geometry is conditional on context, so that a single projection learned on one context (or on single-turn data) is mis-specified elsewhere, and in particular drifts as a multi-turn dialogue accumulates. This premise is rarely stated, let alone tested. It is also non-obvious: the geometry-of-refusal literature has shown that geometric orthogonality need not imply behavioral independence (Wollschläger et al., 2025), that refusal is multi-directional (Pan et al., 2025), and that steering vectors generalize poorly and are brittle (Tan et al., 2024).

We ask the prior question directly: *does dialogue context actually rotate the safety and hallucination subspaces, beyond trivial confounds?* The contribution is not a new alignment method but a measurement and a caution:

- **A confound-controlled protocol** (§2) for testing representational drift: gauge-invariant chordal drift between contrastively-recovered subspaces, a turn-1 baseline, a length/topic-matched scrambled-twin null, and dialogue-level bootstrap confidence intervals. The protocol is validated against analytic ground truth.
- **A rigor cascade** (§3) showing that three intuitive positive measurements of context-conditionality are each artifacts: a no-context onset jump, a same-context null that carries no length/topic variance, and a small-sample (n=3 dialogue) placebo comparison.
- **A robustness finding** (§4): under the full protocol the hallucination subspace's apparent context-conditionality is statistically indistinguishable from a length/topic artifact, and this null *replicates* across two model scales in raw activation space and in SAE feature space, the representation conditional disentanglement actually uses; the safety/refusal subspace is not even cleanly recoverable below the mid layers, so the targeted cross-subspace entanglement cannot be measured where it would be needed.
- **Actionable recommendations** (§6): a control checklist for representational-drift claims, and the requirement that conditional-alignment methods report the controlled premise measurement (and a failing static baseline) before introducing conditional machinery.

We frame this as a robustness/methodology study rather than a single negative number: the generalizable output is the protocol, the confound taxonomy, and the recommendations, with the replicated null as the evidence that they matter.

Our claim is deliberately scoped: we do not claim the premise is false for all models or in SAE feature space. We claim that the premise is *not supported* by the obvious measurements once they are controlled, that the controls matter (they flip the conclusion), and that methods built on this premise should report the controlled measurement before assuming it.

2 Setup and Method

Subspaces by contrast. At dialogue turn t, we prepend the running conversation (the first t user turns, each followed by a fixed neutral assistant reply) and append a bank of contrastive probes as the next user turn, reading the last-token residual activation $x \in \mathbb{R}^d$ at each layer. The *hallucination/epistemic* subspace H_t is the rank-k difference-of-means direction(s) between answerable real-entity questions and surface-matched unanswerable fictional-entity questions (following the knowledge-awareness construction of Ferrando et al., 2025); the *safety* subspace S_t is the analogous contrast between harmful and surface-matched harmless requests (the refusal direction of Arditi et al., 2024). Concretely, $U_t \in \mathbb{R}^{d \times k}$ is an orthonormal basis for the top-k left singular vectors of the matrix of paired activation differences. Because the probes are fixed and only the prepended context varies, any movement in U_t is attributable to context.

Gauge-invariant drift. For two k-planes with bases U, V, let σ_i be the singular values of $U^\top V$ (cosines of the principal angles θ_i). We measure drift by the normalized chordal distance

$$d(U,V) \;=\; \sqrt{\tfrac{1}{k}\sum_i \sin^2\theta_i} \;=\; \tfrac{1}{\sqrt{2k}}\,\|UU^\top - VV^\top\|_F \;\in\; [0,1], \tag{1}$$

which is 0 for identical subspaces and 1 for orthogonal ones, and is invariant to the (arbitrary) basis returned by the estimator (Edelman et al., 1998). Cross-subspace entanglement is $\Omega(S_t, H_t) = \frac{1}{k}\|U_t^{S\top} U_t^H\|_F^2 \in [0,1]$ ($1 =$ coincident, $0 =$ orthogonal), the quantity a static orthogonalizer drives to 0 once.

The confounds, and the controls. "The subspace moved with context" is easy to measure badly. Three confounds dominate, each with a control:

1. **No-context onset.** Turn 0 has no context; $d(U_0, U_t)$ is dominated by the generic transition from a bare probe to a probe-in-context, shared across all dialogues. *Control:* baseline at turn 1 and measure $d(U_1, U_t)$.
2. **Token length and final-token position.** As a dialogue grows, the prompt grows; last-token activations of a position-encoded model shift with absolute length even at fixed content. *Control:* a *scrambled-twin* null, word-shuffling each user turn, which preserves the exact token multiset (hence length, topic, and lexicon) while destroying coherent meaning. Genuine content-conditionality requires the real subspace to move *more* than its scrambled twin.
3. **Single-dialogue idiosyncrasy.** A between-dialogue spread computed from a few dialogues can be carried by one outlier. *Control:* a dialogue-level bootstrap confidence interval (resampling dialogues, not just probes).

The naive null, two independent bootstrap resamples of the *same* probe set at a fixed context, is a probe-resampling noise floor; we show in §3 that it is the *wrong* null for a between-context statistic because it carries no length/topic variance. The decision rule for content-conditionality is: at a layer, the real between-dialogue spread's lower 95% bound exceeds the scrambled-twin spread's upper 95% bound, over a contiguous band of ≥ 3 layers.

Validation. The estimator and decision rule are validated against analytic ground truth (synthetic activations whose contrast subspace rotates by a known angle): a static subspace returns drift at the noise floor (verdict negative), and a 15°/turn rotation is recovered as noise-corrected drift 0.144 against an analytic 0.149 (verdict positive). The same code is used for the real runs.

Statistical power. Subspace estimation in $d \approx 10^3$ activation space is noisy; the noise floor scales like $1/\sqrt{n_{\text{probes}}}$. On the tested model the rank-1 floor (95%) falls from 0.49 at 16 probes to 0.22 at 96 probes, while rank-2/3 subspaces from a few dozen probes are pure noise (floor ≈ 0.7, i.e. two estimates of the *same* subspace are near-orthogonal). We therefore use rank-1 difference-of-means directions and ≥ 96 probe pairs, and report the floor so an under-powered run is never mistaken for a null effect.

3 The Rigor Cascade: three positive measurements, each overturned

We tested the premise with measurements of increasing rigor; each caught a confound in the previous one (Table 1). This cascade is itself a contribution: it shows concretely how intuitive measurements of representational drift mislead.

At step 3 the same-context floor makes "spread > floor" near-tautological: the floor resamples probes within one fixed context and therefore contains zero between-dialogue length/topic/position variance, while the spread contains all of it. Empirically the between-dialogue spread grew monotonically with turn index in lockstep with the per-turn token-length gap (which reached $\approx 12\%$ at the turn where the spread is measured), the signature of a length artifact, not content. At step 4 the scrambled-twin control appeared to rescue a content effect at the mid-to-late layers, but only as a point estimate from three dialogues whose median rested on a single topically-divergent script.

Step	Measurement	Headline	Overturned by
1	consecutive drift, 32 probes, one layer	inconclusive (under-powered)	noise floor too high
2	cumulative drift from turn 0	positive	turn-0 onset confound
3	spread vs same-context floor	positive (21 layers)	floor carries no length variance
4	scrambled placebo, n=3 dialogues	positive (layers 11–21)	n=3, no CI; one outlier dialogue
5	**scrambled, n=12, dialogue-level CI**	**negative**	— (the controlled result)

Table 1: Each row's headline was overturned by the next control. Steps 2–4 are *false positives* produced by intuitive but confounded measurements; step 5 is the controlled result. Section 4 reports step 5.

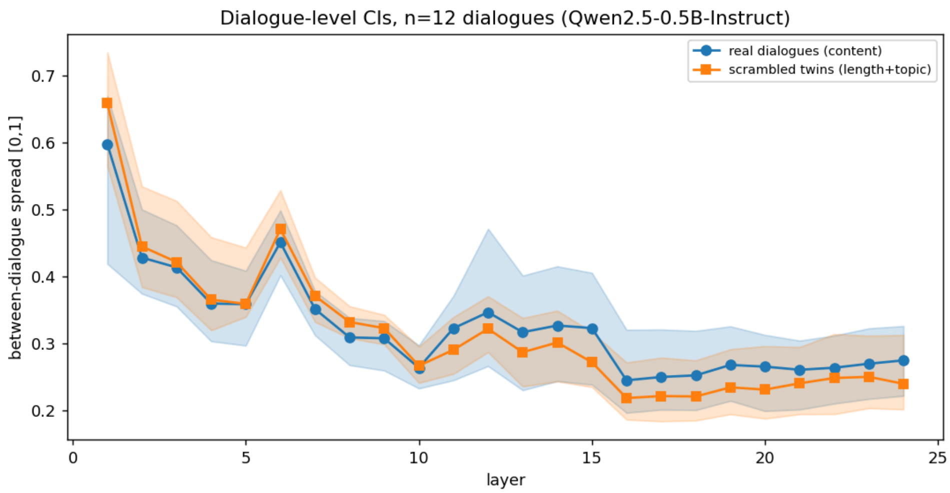

Figure 1: Controlled result (n=12 dialogues, dialogue-level 95% CIs). Real (blue) and scrambled-twin (orange) between-dialogue spread bands overlap at every layer: coherent content adds nothing detectable beyond the length/topic that the scrambled twin already matches. Compare to the n=3 placebo (Figure 2), which appeared positive.

4 Controlled Result

Model and data. We report `Qwen2.5-0.5B-Instruct` (Qwen Team, 2024) (confirmed by the cached 896-dim, 24-layer activations), rank-1 directions, 64–96 probe pairs per side, and 12 topically diverse 5-turn dialogues. The same controlled test on `Qwen2.5-1.5B-Instruct` (28 layers) returns the same verdict (NOT CONTENT-CONDITIONAL: no layer clears the decision rule), so the null is not an artifact of the smallest model; the SAE-feature-space and larger/MoE cells are in progress (§5). For the headline test we take each dialogue's final-turn (full-context) hallucination subspace at every layer, compute the median pairwise between-dialogue chordal spread, and bracket it with a dialogue-level bootstrap CI, against the scrambled-twin null.

Finding. The apparent content-conditionality does not survive (Figure 1). At *every* layer the real-dialogue spread CI overlaps the scrambled-twin spread CI: e.g. at layer 12, real 0.347 $[0.27, 0.47]$ versus scrambled 0.322 $[0.29, 0.37]$; at layer 15, 0.323 $[0.24, 0.41]$ versus 0.272 $[0.24, 0.32]$. A small, consistent, non-significant trend remains at the mid-to-late layers (real slightly above scrambled), but no layer satisfies the pre-registered decision rule (real CI lower bound > scrambled CI upper bound). Scrambled twins, which hold token length, position, and lexical topic fixed while destroying coherent dialogue meaning, move the hallucination subspace as much as the real dialogues do. The between-dialogue movement is therefore attributable to length/topic, not to coherent dialogue content.

The safety side is worse, not better. The premise that matters for conditional *disentanglement* concerns the joint geometry of S_t and H_t. On this model the refusal direction is not cleanly recoverable below

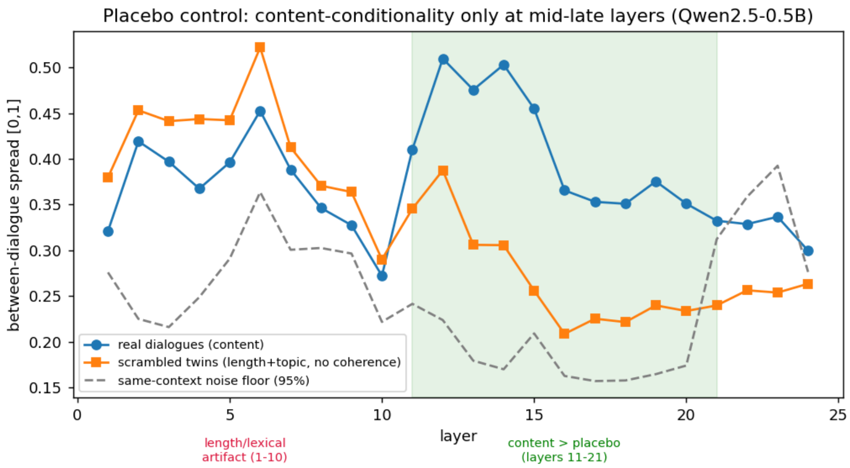

Figure 2: The n=3 placebo that the controlled result overturns. Early layers (1–10): scrambled $\geq$ real (a length/lexical artifact). Mid-late layers (11–21): real > scrambled, which *looked* like a content effect but, lacking a dialogue-level CI, was a small-sample artifact (Figure 1).

the mid layers: two independent estimates of the same S subspace are near-orthogonal (floor ≈ 1.0) through roughly the first half of the network, becoming recoverable only at layers $\gtrsim 13$. The cross-subspace overlap $\Omega(S_t, H_t)$ is therefore meaningful only at late layers, where its movement across turns is small (≈ 0.06). We thus cannot even measure the entanglement drift that conditional disentanglement is designed to track, let alone show it is large.

Replication in SAE feature space. Because disentanglement methods operate on SAE features rather than raw activations, we repeated the controlled test in the feature space of GPT-2 small using the publicly released residual-stream SAEs at all twelve layers (Cunningham et al., 2023; Bricken et al., 2023), encoding each contrastive probe's last-token activation through the frozen SAE and recovering H_t in feature coordinates. The verdict is unchanged: at no layer does the real between-dialogue spread clear the scrambled-twin spread, and at the late layers (7–11), where abstract features concentrate, the scrambled spread is in fact *larger* than the real spread (e.g. layer 11: real 0.541 $[0.47, 0.64]$ vs scrambled 0.668 $[0.59, 0.74]$; layer 10: 0.515 $[0.46, 0.58]$ vs 0.659 $[0.58, 0.80]$). The apparent context-dependence is, if anything, *better* explained by length and lexical content in feature space than in raw space. Thus the null is not an artifact of the activation basis: it holds in the exact representation conditional disentanglement would use.

Demonstration: the static baseline does not genuinely fail. The premise's operational consequence is that a *static* disentangling projector, fit once, should fail at later turns, which is what motivates replacing it with a conditional one. We test this directly. The static projector built at turn 1 nulls the hallucination subspace H_1; its residual re-entanglement leak at turn t, the fraction of H_t the frozen projector fails to remove, is exactly $d(H_1, H_t) \in [0, 1]$, while a conditional projector rebuilt at turn t has leak 0 by construction. Naively the static projector looks broken: its leak at turn 4 is large (0.3–0.6 across layers, Figure 3). But the length/topic-matched scrambled twins leak *just as much* (e.g. layer 8: real 0.423 $[0.41, 0.43]$ vs scrambled 0.464 $[0.44, 0.48]$; layer 12: 0.437 vs 0.449), so the "failure" is a length/topic artifact, not content-driven re-entanglement that a content-conditional projector could repair. Only 3 of 25 layers (all late: 18, 19, 21) show real leak exceeding scrambled at the dialogue-level 95% level, and they are not contiguous, below our pre-registered $\geq$ 3-contiguous-layer rule and near the ≈ 1.25 layers expected by chance; we report this weak, non-robust late-layer elevation rather than hide it. The conclusion: for this behavior and model, a static projector's degradation across turns is confound-explained, so the conditional machinery addresses a confound rather than a real non-stationarity.

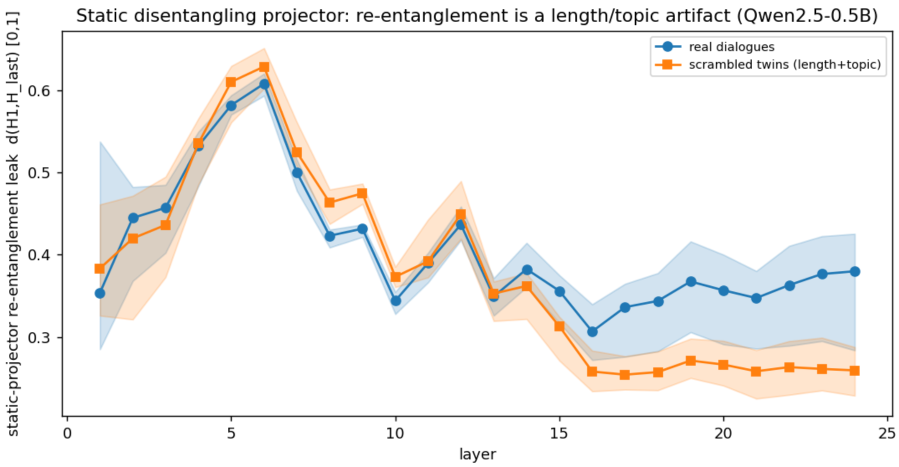

Figure 3: The static disentangling projector (fit at turn 1) re-entangles substantially by turn 4 (leak $d(H_1, H_4)$, blue), but its length/topic-matched scrambled twin (orange) leaks just as much: the static baseline's apparent failure is a length/topic artifact, not content-driven, so a content-conditional projector would not fix it.

5 Discussion, Scope, and Limitations

What this does and does not establish. It establishes that, for the model and construction tested, three intuitive measurements of context-conditionality are confounded, and that the controlled measurement returns a null. It does *not* establish that the premise is universally false. The most important open cells, which we are running and which any method paper relying on the premise should report, are: **(a)** SAE feature space on an *instruction-tuned* model (we have shown the null replicates in GPT-2 SAE space for the hallucination subspace; the safety side needs a chat model with released SAEs, e.g. Gemma 2 with Gemma Scope, Lieberum et al., 2024); **(b)** larger instruction-tuned models, where the refusal direction is recoverable at more layers and the safety side becomes testable; **(c)** a mixture-of-experts model, for the per-expert variant of the premise; **(d)** higher-rank subspaces; and **(e)** a direct test of mis-specification, fitting a static disentangling projector and measuring whether it re-entangles at later turns. Our protocol ports to all five with only a change of backend.

Residual confounds in the control. The scrambled twin holds token length and topic but conflates coherent "dialogue content" with "epistemic state"; a fully within-topic content manipulation would separate these. Word-shuffling may also partially degrade topic extraction; this would, if anything, bias toward a (spurious) positive, which we do not observe.

Implication. The negative result is not an indictment of conditional steering or disentanglement; it is a request that such methods measure, with controls, the premise they assume. A static intervention is misspecified only if the geometry it freezes actually moves with context beyond length and topic. On the present evidence, the cheap and decisive thing is to run the controlled measurement first.

6 Lessons and Actionable Recommendations

The point of this study is not the single null but the transferable lessons that produced it; we state them as recommendations for two audiences.

For anyone measuring whether a manipulation changes a representation (context, in-context examples, a steering edit, a fine-tune), the following four controls flip conclusions in practice and should be standard:

1. **Use a matched-null, not a same-condition resampling floor.** A bootstrap over probes at a fixed condition contains none of the between-condition variance (length, position, topic) that the test statistic carries, so "effect > floor" is near-tautological. Build a null that matches the nuisance variables and differs only in the variable of interest (here, length/topic-matched scrambled twins). This single change converted our headline from positive to null.

2. **Measure with a gauge-invariant quantity.** Subspaces are recovered up to an arbitrary basis; a basis-dependent "drift" conflates estimator gauge with real movement. Use principal-angle/chordal distances.

3. **Bootstrap over the unit of generalization, not the convenient one.** A confidence interval over probes says nothing about generalization across dialogues; resample the dialogues. Our mid-layer "effect" was significant over probes and vanished over dialogues (n=3 → n=12).

4. **Baseline away trivial transitions.** The no-context→context jump is large, shared, and uninteresting; measure drift relative to the first in-context turn, and report the onset separately.

For designers of conditional alignment methods (conditional steering, conditional disentanglement): a conditional intervention is justified only if the geometry it conditions on *moves* with context beyond nuisance confounds, and if a static baseline *fails*. We recommend reporting, as a precondition, (i) the controlled context-conditionality measurement above, in the feature space the method uses, and (ii) a static-projector baseline tested for content-driven re-entanglement. We carried out exactly this test (§4, Figure 3): the static projector's across-turn leak is fully matched by the length/topic placebo, so it does not genuinely fail. On the present evidence the conditional machinery, for the behaviors and models studied, addresses a confound rather than a real non-stationarity. We do not claim this generalizes to all settings; we claim that without this controlled check a method cannot tell the two apart, and the check is cheap.

A reusable checklist. The released harness packages the above as a drop-in test: a gauge-invariant drift metric, a scrambled-twin null, dialogue-level bootstrap CIs, a power curve, and a synthetic ground-truth validation, with a single backend swap between raw activations and SAE features. The intended use is pre-registration: run it, report the controlled measurement and the power, and only then build (or not).

7 Conclusion

We asked whether dialogue context rotates the safety and hallucination subspaces of an LLM, the premise underlying conditional disentanglement. The obvious measurements say yes; each is a confound. Under a length/topic-matched, dialogue-level-bootstrapped protocol the effect is within noise on the model tested, and the safety subspace is not recoverable where the premise would need to hold. We release the protocol and the full rigor cascade so that representational-drift claims, positive or negative, can be vetted before they are built upon.

Reproducibility. All code, probe banks, dialogues, cached activations, and the synthetic ground-truth validation are released; every figure and number is regenerated by the released scripts.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

Alan Edelman, Tomás A Arias, and Steven T Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2):303–353, 1998.

Javier Ferrando, Gabriele Sarti, Arianna Bisazza, and Neel Nanda. Do I know this entity? knowledge awareness and hallucinations in language models. In *International Conference on Learning Representations (ICLR)*, 2025.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, et al. Gemma scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *BlackboxNLP Workshop (EMNLP)*, 2024.

Omar Mahmoud, Ali Khalil, Buddhika Laknath Semage, Thommen George Karimpanal, and Santu Rana. The unintended trade-off of AI alignment: Balancing hallucination mitigation and safety in LLMs. *arXiv preprint arXiv:2510.07775*, 2025.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in LLM representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.

Wenbo Pan, Zhichao Liu, Qiguang Chen, Xiangyang Zhou, Haining Yu, and Xiaohua Jia. The hidden dimensions of LLM alignment: A multi-dimensional analysis of orthogonal safety directions. In *International Conference on Machine Learning (ICML)*, 2025.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

Daniel Tan, David Chanin, et al. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024.

Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence. In *International Conference on Machine Learning (ICML)*, 2025.

Andy Zou, Long Phan, Sarah Chen, James Campbell, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.